Data Visualization Pitch

Consumer Behavior in Istanbul's Shopping Malls

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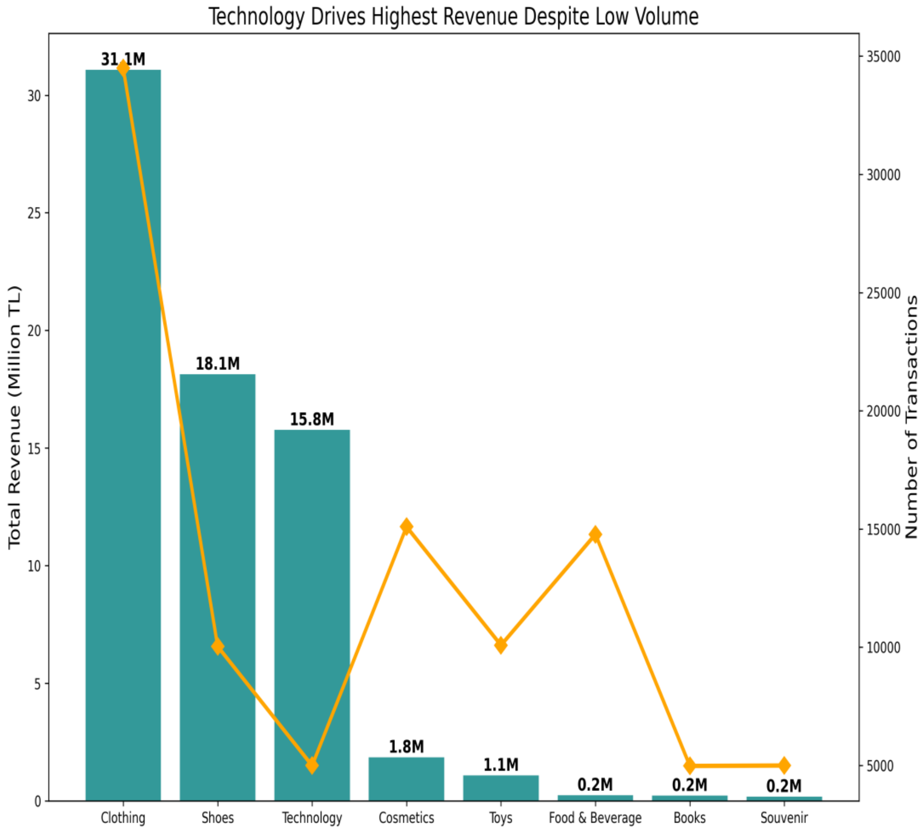
Student Individual Assignment



Unveiling Istanbul Shopping Trends: High-Value Categories and Mall Dominance (2021–2023)

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Research questions

What are the key drivers and patterns of consumer spending in Istanbul's shopping malls (2021–2023)?

- Which product categories generate the highest revenue, and how do they compare to transaction volume?
- Which shopping malls perform best in terms of sales volume and revenue?
- Is there a seasonal or holiday trend in shopping activity and spending?

About Data

Data link → [here]

The dataset chosen for this analysis is “customer_shopping_data.csv,” which provides detailed records of 99,457 transactions from 10 shopping malls in Istanbul between 2021 and 2023. The data includes several attributes such as invoice numbers, customer IDs, gender, age, item categories (8 types), quantity, price (in TL), payment methods, invoice dates, and shopping malls. The data offers insights into consumer shopping behaviors, including real-world patterns such as holiday peaks.

While searching for a dataset on shopping trends in Istanbul, I found this dataset on Kaggle by searching "retail shopping trends Istanbul." The data collection itself was straightforward, and no major issues were encountered. I did consider alternative datasets, such as Eurostat tourism data, which could provide broader insights. However, this dataset stood out due to its focus on retail behavior within Istanbul's malls and its granular data on consumer behavior.

The dataset is licensed under the Creative Commons BY-SA 4.0 license, which allows for free use, redistribution, and adaptation, as long as proper attribution is given. There are no specific restrictions noted for this dataset, and it is open for analysis.

Concerning the quality and completeness of the data, while the dataset is largely complete and includes accurate transaction details, its scope is confined to 10 shopping malls in Istanbul. This regional limitation may restrict the findings’ applicability to broader retail markets in Turkey. Furthermore, all values are recorded in Turkish Lira (TL), and no currency conversion is applied, which might pose challenges for comparisons over time or across regions. Despite these gaps, the data's quality remains high for the intended analysis.

Regarding data cleaning, minimal operations were needed. I transformed the invoice_date column into a datetime format using Python to facilitate time-based analysis, checked for duplicate records (none were found), and added a new column to calculate the total price for each transaction (quantity × price) for revenue analysis.

Methodology

Data Collection: Downloaded CSV from Kaggle. No scraping/APIs needed.

Data Processing & Cleaning: Used Python with pandas in Jupyter notebook. Loaded CSV, handled date formats (pd.to_datetime with dayfirst=True), verified no missing values, added calculated column 'total_price' = quantity * price. No duplicates dropped.

Data Transformation: Aggregated revenue (groupby 'category'/'shopping_mall' on total_price.sum()), created monthly bins ('month_year' = dt.strftime('%Y-%m')), binned age groups (pd.cut for 18-30, etc.), sorted for rankings. Normalized to million TL for readability.

Use of AI Tools: Grok (xAI) for code suggestions and initial aggregations (e.g., groupby queries); verified manually in Jupyter.

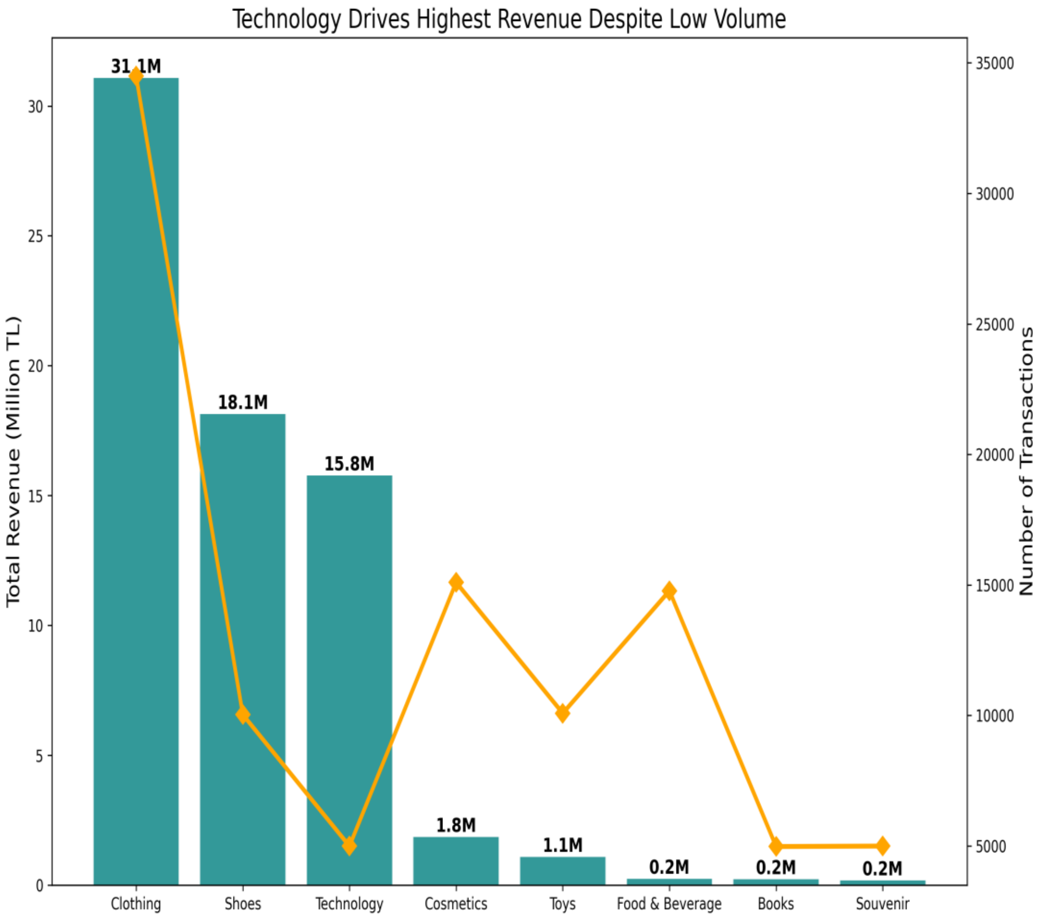
Analytical Techniques: Descriptive stats (sum/mean via groupby), trend analysis (monthly sorting), comparative analysis (revenue vs count), binning (age groups). Outlier detection: No major (prices within expected range).

Reproducibility: Fully replicable. Code shared in Jupyter notebook (attached on GitHub link); steps: Download CSV from Kaggle, run pandas groupby for aggregates, export to CSV for viz tools.

Insights from the Data

- **Categories drive dramatic revenue gaps:** Clothing generates highest volume (34,509 transactions) but Technology leads revenue at 57.86 million TL despite only 4,997 transactions – high-value items dominate profits.
- **Gender patterns:** Females account for 59% of transactions and 60% of revenue (150.21 million TL vs males' 101.3 million TL), with women favoring Cosmetics/Clothing, men Technology/Shoes.
- **Age groups:** Young adults (18-30) contribute most (58.65 million TL, ~25% total), but older groups (60-70) have higher average spend per transaction (~1,200 TL) on luxury like Technology.
- **Malls show concentration:** Mall of Istanbul and Kanyon together generate 101.45 million TL (~40% total revenue), with clear drop-off to others – indicating market dominance.
- **Time trends:** Monthly revenue peaks in July/October 2021-2022 (over 10 million TL), dropping sharply in 2023-03 (2.51 million TL) – strong holiday/seasonal effects (e.g., back-to-school, New Year).
- **Analysis operations:** Calculated total_price, groupby sum/mean/count, datetime parsing for trends, sorting for rankings. Most interesting for viz: Category revenue vs volume (contrast), mall rankings (dominance).

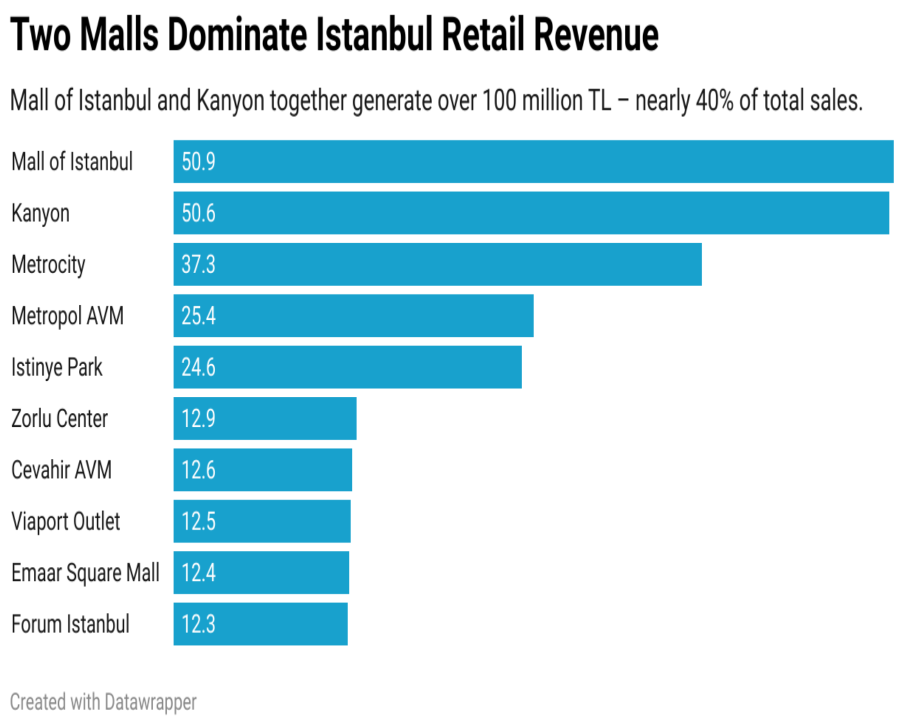
Category Revenue vs Volume



This visualization illustrates how different categories in Istanbul shopping malls contribute to revenue and transaction volume. For instance, clothing has the highest transaction volume but a relatively lower revenue compared to technology. Technology, despite having fewer sales, generates the highest revenue. This indicates that consumers tend to spend more per transaction in technology than in clothing, which is a significant pattern to consider when analyzing spending behavior.

The **bar chart** on the left displays the total revenue for each category (in million TL), while the **orange line** represents the number of transactions. Technology stands out as a high-revenue but low-transaction category, while clothing dominates the volume side but generates lesser total revenue. The insights derived from this visualization can help businesses understand how to allocate resources for maximum profitability. For instance, while focusing on high-volume categories, promotions or strategies aimed at technology could yield higher revenues per transaction.

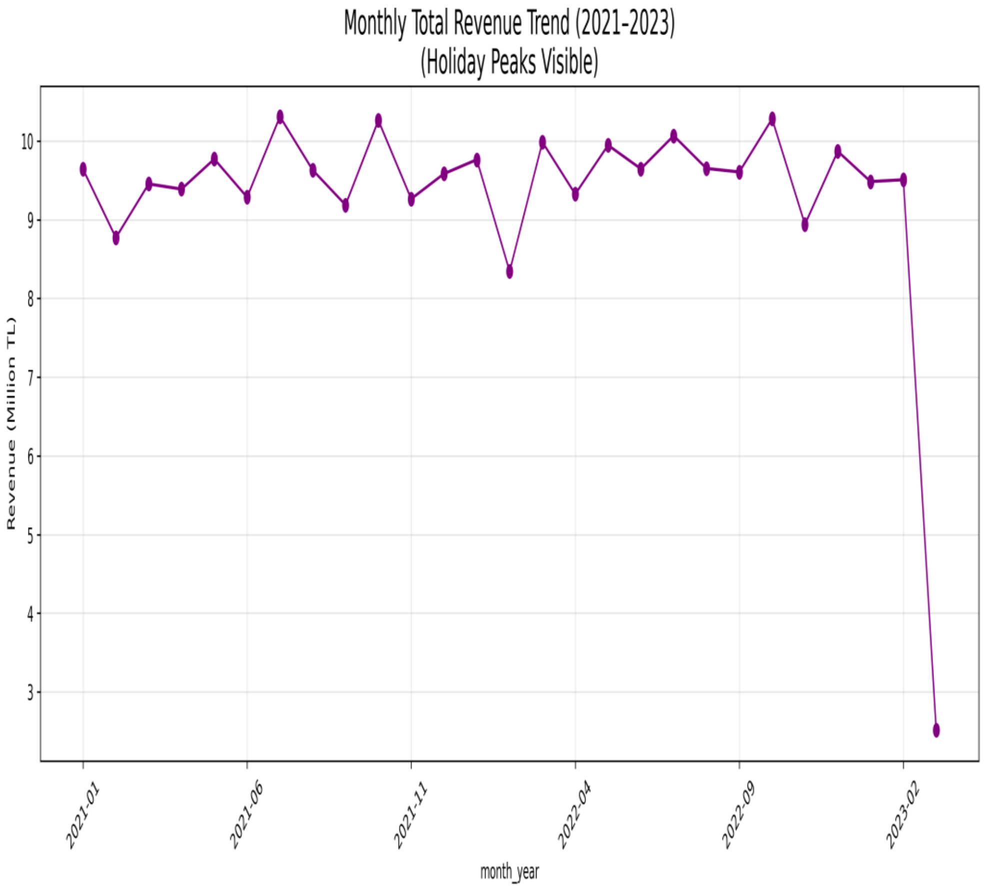
Two Malls Dominate Istanbul Retail Revenue



- The bar chart clearly shows that Mall of Istanbul and Kanyon generate significantly more revenue compared to others like Metrocity, Metropol AVM, and Istinye Park. This data could help guide future investments or expansion plans, as businesses may want to focus more resources on the top-performing malls while also considering how smaller malls can boost their performance.

- Furthermore, the Mall of Istanbul and Kanyon together account for nearly 40% of the total revenue. This concentration of revenue in just a few malls suggests that these two are market leaders, and smaller malls should explore ways to increase their market share, perhaps through targeted promotions or events.

Monthly Revenue Trend



- The third visualization illustrates monthly revenue trends from 2021 to 2023, showing clear peaks during certain months. These peaks correspond with major shopping periods, such as holidays and end-of-season sales. For instance, the data shows a spike in revenue around New Year's and back-to-school periods, suggesting strong seasonality in consumer spending."
- To be exact, it captures the fluctuations in monthly revenue, with highlighted peaks during certain periods. This seasonal variation in shopping behavior is important for businesses to understand, as it suggests when to expect high sales and when to prepare for quieter months.

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